

Module 2 Reflective Journal

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This project was my first time using Google Colab and working through a machine learning notebook. When I first opened Colab, it was very intimidating because I did not even know Google offered a service like this or what it was used for. There were a lot of unfamiliar tools and options on the screen, and that made me nervous because I did not understand what they did or how they could be useful.

Colab started to feel less intimidating once I began running the code. It was simple to click the play button for each cell, and having the instructions close to the code helped me understand what I was supposed to do. I realized that I did not need to understand every tool on the screen before I could start using the notebook. Working through the project one step at a time made the environment feel much more manageable.

I had also never used pandas, NumPy, or matplotlib before this project. It is still difficult for me to remember exactly what each library does, but I became interested in how they all worked together. One tool helped organize the data, another performed calculations, and another turned the data into visualizations. Seeing them used together helped me understand that machine learning projects use different tools for different parts of the process.

The visualization section was another important part of the lab for me. When I first saw the scatter plot, I felt like I had no idea what I was looking at or what the points meant. As I continued through the lab and reached the bar graph, the data started to make more sense. I could clearly see the three flower species and that each one had 50 samples. This helped me understand how graphs can make a large amount of data easier to understand than looking only at rows of numbers.

This project also changed the way I think about machine learning. Before this lab, I did not fully understand how much work could be done with data before even building a model. We were able to organize the data, calculate statistics, create graphs, compare different groups, and look for patterns. That made me curious about what else can be discovered, predicted, or understood by using different datasets and machine learning tools.

Another benefit of this project is that I am currently taking a Python class. Seeing Python used in a real machine learning project made me feel more confident in that class. Instead of only seeing small practice problems, I was able to see how Python can actually be used to work with data, perform calculations, and create visualizations. That made the code feel more meaningful and gave me a better idea of how Python can be used outside of basic exercises.

Overall, I feel much more comfortable using Google Colab now than I did at the beginning of the project. I still have a lot to learn about pandas, NumPy, matplotlib, and machine learning, but the tools no longer feel as intimidating. This lab helped me see how coding, data, statistics, and visualization can all work together, and it made me more interested in learning what else can be done with data.